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# The Role of Digital Twin for Space and Remote Terrestrial Construction

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**Correspondence:** Ehsan Noroozinejad Farsangi ([ehsan.noroozinejad@westernsydney.edu.au](mailto:ehsan.noroozinejad@westernsydney.edu.au))**Received:** 20 May 2025 | **Revised:** 18 January 2026 | **Accepted:** 27 January 2026**Keywords:** digital twin | ESA | extreme-environment construction | JAXA | NASA | predictive maintenance | remote infrastructure | space habitats | SpaceX | swarm robotics

## ABSTRACT

Digital twin technology, continuously updated, high-fidelity virtual replicas of physical assets, has matured rapidly in aerospace and is now poised to transform construction in extreme environments on Earth and beyond. This Opinion Article synthesises recent research, global case studies, and policy developments to argue that digital twins are pivotal for planning, autonomous execution, and predictive infrastructure maintenance on the Moon, Mars, the deep ocean, polar regions, and remote deserts. By coupling real-time sensor data with physics-based and AI-driven simulation, twins enable risk-free prototyping, swarm-robot coordination, and lifecycle resilience while reducing environmental footprints. We highlight strategic implications, data-governance frameworks, workforce upskilling, and federated twin ecosystems and forecast emerging trends such as AI-enhanced autonomy, XR interfaces, and smart-material feedback loops. The article aims to inform multidisciplinary researchers, engineers, and policymakers by advocating the early adoption of digital twin standards to accelerate sustainable resilient construction in the most challenging frontiers.

## 1 | Introduction

Extreme environments, from the vacuum of space to the depths of the ocean and the frozen expanse of the Arctic, pose unprecedented challenges to construction and infrastructure development. Building and maintaining structures in these settings is fraught with difficulties: human access is limited or dangerous, conditions are unpredictable, real-time communication may be delayed, and mistakes or failures can be costly or life-threatening. As humanity prepares to establish a sustained presence on the Moon and Mars and expands operations in remote terrestrial regions, there is a pressing need for technologies to mitigate these challenges. Digital twin technology has rapidly gained attention as a potential game-changer in this context. A digital twin is essentially a living digital replica of a physical asset or system, kept continuously in sync with the real world via sensors, data streams, and simulations. By providing an up-to-date virtual mirror of physical reality, digital twins

enable engineers and operators to monitor conditions in inaccessible locations, predict issues before they occur, and even rehearse complex operations in silico, all of which are invaluable for extreme environment construction [1–10].

The concept of the digital twin originated in aerospace and defence contexts and is deeply rooted in NASA's early experiences. During the Apollo programme of the 1960s–70s, NASA developed what could be viewed as the first digital twin: a 'living model' of the Apollo spacecraft that mission control used to diagnose and solve problems remotely. In the Apollo 13 crisis of 1970, for example, simulators and physical models augmented with real telemetry were used to replicate the damaged spacecraft's condition, allowing engineers on Earth to devise a successful rescue procedure. This paved the way for the formal concept, decades later, NASA's John Vickers coined the term 'digital twin' around 2010 as the agency embraced model-based engineering for next-generation programs. Since then, digital

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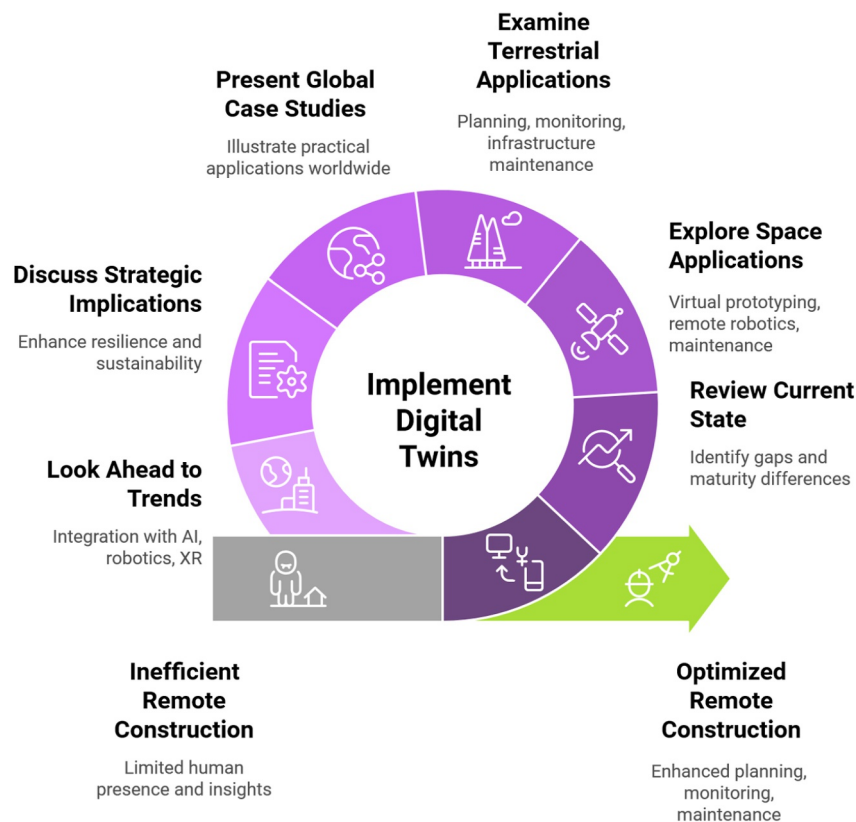
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twins have evolved with advances in computing, sensors (Internet of Things), and AI and have been adopted far beyond aerospace: they now drive innovation in manufacturing, urban planning, energy, healthcare, and more. Crucially, the underlying value of digital twins lies in their ability to continuously integrate data (from sensors, surveys, etc.), simulate behaviours under various scenarios, and inform decision-making in real-time [11–15].

For the construction industry, digital twins represent an extension of earlier digital modelling techniques such as Building Information Modelling (BIM) and computer-aided design, adding real-time data and operational analytics to static models. This dynamic data-rich modelling is particularly relevant for projects in extreme environments where traditional on-site oversight is infeasible. By creating a high-fidelity virtual construction site or infrastructure asset, project teams can ‘be present’ virtually, analysing progress or problems from thousands of kilometres away. For instance, engineers could supervise an autonomous construction rover on the lunar surface via its digital twin, receiving continuous feedback on the rover’s performance and the surrounding environment, and even intervening or re-planning tasks through that virtual interface. Similarly, a digital twin of an Arctic pipeline can display real-time stresses, ground temperature, and structural integrity, alerting operators to emerging risks such as permafrost thaw or material fatigue long before a failure occurs [1, 16–19].

This opinion article delves into the role of digital twins in enabling construction and infrastructure development in space and other remote or extreme terrestrial environments. We structure the discussion as follows (Figure 1):

- We begin with a clear definition of digital twins and the fundamental principles that make them powerful in establishing common understanding across our multidisciplinary audience.
- We review the current state of digital twin technology in both the construction sector and the space sector, highlighting gaps and differences in maturity between these domains.
- We explore applications of digital twins for space construction, encompassing lunar, Martian, and orbital projects, including virtual prototyping/design, remote robotic construction, and predictive maintenance of space assets.
- We examine applications in remote terrestrial construction, such as projects in the Arctic, underwater, and deserts, focussing on how digital twins aid in planning, monitoring, and maintaining infrastructure where human presence is limited.
- We present global case studies from public agencies and private companies to illustrate how theory is being put into practice around the world.
- We discuss the strategic and policy implications, arguing that digital twins can enhance infrastructure resilience and sustainability but also require policy support in terms of standards, investment, and collaboration frameworks.
- Finally, we look ahead to future trends, including integration with AI and machine learning, robotics and autonomous swarms, extended reality interfaces, smart materials, and interconnected digital twin ecosystems poised to shape the next generation of construction technology.



**FIGURE 1** | Schematic presentation of the article outline. (Developed by the author for this paper).

Throughout, we aim to maintain a style that is rigorous and evidence-based yet accessible to policymakers and practitioners, emphasising the transformative potential of digital twins for meeting the engineering challenges of tomorrow's 'extreme' projects.

## 1.1 | Definition and Foundational Principles of Digital Twins

A digital twin can be defined as a virtual representation of a physical object, system, or process that is updated continuously with real-world data and used to inform decisions through simulation and analysis. In essence, it is a computerised clone of a real entity, mirroring that entity's state and behaviour in near real-time. Several authoritative definitions converge on the key characteristics of digital twins. For example, a 2024 National Academies report defines a digital twin as '*a set of virtual information constructs that mimics the structure, context, and behaviour of a natural or engineered system... is dynamically updated with data from its physical twin, has predictive capability, and informs decisions that realise value*,' emphasising that bidirectional interaction between the physical and virtual system is central. The Digital Twin Consortium similarly describes it as '*an integrated data-driven virtual representation of real-world entities and processes, with synchronised interaction at a specified frequency and fidelity*,' highlighting the continuous synchronisation of data. In simpler terms, the digital twin stays in lockstep with its physical counterpart: as the physical asset changes or operates, sensors and data feeds update the digital model; conversely, analyses or control actions from the digital side can be applied back to the physical side [13, 20–22].

Foundational principles that underpin digital twins are depicted in Figure 2.

- **Real-Time Data Synchronisation:** Digital twins rely on data streams from sensors (e.g. IoT devices embedded in equipment, structural health monitors, environmental sensors) to ensure the virtual model reflects current reality. This could be high-frequency telemetry from a machine or periodic updates on ecological conditions. The constant data flow enables the twin to be a live mirror of the system's state [23–25].
- **Simulation and Predictive Modelling:** A digital twin is not a static model; it is usually coupled with physics-based models, simulations, or AI algorithms that can predict future behaviour or test scenarios. By leveraging high-performance computing, digital twins can answer 'what if' questions, such as how a lunar habitat will respond to a meteoroid strike or how an underwater pipeline would behave in a severe storm without risking the real asset. This predictive capability allows proactive interventions (maintenance, design changes, contingency planning) before real problems occur [26–30].
- **Bidirectional Feedback and Control:** Importantly, digital twins are connected to their physical twins in a feedback loop. Not only does data flow from the physical to the digital (to update the model) but insights and commands can flow from digital to physical. In practice, this means an

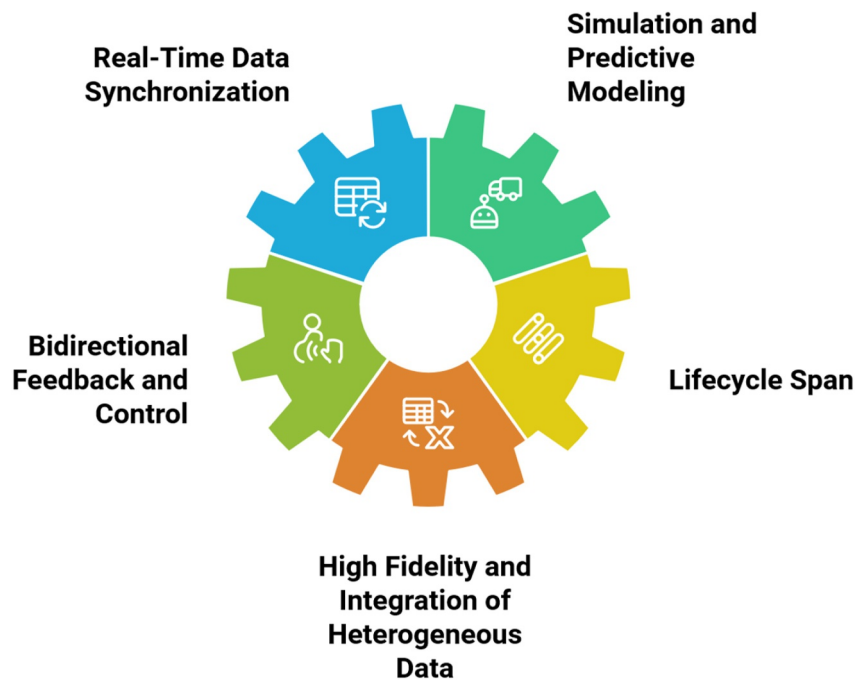
operator could use the digital twin to send optimised control inputs to a robot or adjust the operating parameters of remote equipment. For example, if a digital twin detects via simulation that a lunar rover's motor is overheating, it might trigger a command to reduce speed or suggest a cooling pause, thus directly influencing the physical operation [25, 31, 32].

- **Lifecycle Span:** A true digital twin often spans the entire lifecycle of an asset from design and construction through operations and maintenance to eventual decommissioning. In the design phase, the twin (sometimes called a virtual prototype) allows testing and optimisation. During construction, it can track progress and quality. In operations, it monitors performance and degradation. And at end-of-life, it can assist in safely decommissioning or repurposing assets. This lifecycle perspective is crucial for long-term infrastructure such as space habitats or remote installations, ensuring continuity of knowledge and data [16, 26, 27, 32].
- **High Fidelity and Integration of Heterogeneous Data:** Digital twins integrate data from multiple sources and disciplines, such as structural, mechanical, electrical, environmental, and even human factors data. The fidelity can range from coarse (for strategic planning) to ultra-fine (for detailed engineering). For instance, a digital twin of a spacecraft might encompass its 3D CAD geometry, sensor readings from its subsystems, orbital dynamics models, and even software states. The richer and more accurate this integrated model, the more value it provides. In construction, this means merging BIM data, schedule (4D) data, cost (5D) data, and real-time sensor data into one cohesive virtual space [16, 21, 28, 33].

Together, these principles allow digital twins to function as 'living' models of their physical counterparts, supporting what NASA researchers have called a shift towards '*living models*' or '*living digital simulations*' that evolve alongside the system [34]. The idea of continuously living models marks a departure from older practice where simulations were one-off or sporadic. Instead, the digital twin is constantly running and continuously updated, providing an ongoing decision-support tool.

It is crucial to differentiate digital twins from related concepts. A digital twin is more than a simple CAD model or a static 3D visualisation; it is also more than just raw sensor data. Instead, it is the fusion of model + data + analytics in a closed loop. In the construction context, one can think of a digital twin as *BIM plus live data plus intelligence* [35]. Similarly, terms such as digital shadow or digital model sometimes refer to subsets of this idea. A digital shadow might imply a one-way data flow (physical to digital only), lacking interactive, predictive elements. A fully realised digital twin, by contrast, has two-way interactivity and autonomous updating.

From a systems engineering perspective, digital twins are an implementation of cyber-physical systems. They tie together the physical domain and the cyber domain, requiring robust system integration. Key enabling technologies include IoT sensor networks (to gather data), cloud and edge computing (to process data and host models), and connectivity (to transmit data,



**FIGURE 2** | Foundation of a digital twin. (Developed by the author for this paper).

especially challenging in remote environments with limited bandwidth or latency, such as deep space). Advances in machine learning have also enhanced digital twins by allowing data-driven model updates and anomaly detection, complementing physics-based approaches.

A significant technical challenge for digital twin systems in extreme environments is data synchronisation under constrained connectivity, such as deep space, polar regions, or remote terrestrial sites with limited bandwidth or intermittent links. Traditional real-time streaming of high-frequency sensor data is often infeasible in such contexts. To address this, systems adopt edge computing and model-informed buffering approaches: raw sensor data are pre-processed locally to extract essential features and anomalies, and only compressed or high-value summaries are transmitted to the remote twin, whereas less critical data are stored and forwarded when connectivity permits. This reduces bandwidth demand while preserving the twin's operational fidelity. Concurrently, predictive models, including physics-based simulations and machine learning estimators, fill in gaps during communication blackouts, allowing the twin to maintain an up-to-date representation based on the last known state and expected dynamics. Literature indicates that this hybrid approach, combining local processing, prioritised transmission, and predictive interpolation, enables digital twins to function effectively despite low and variable bandwidth conditions [36, 37]. Additionally, federated and distributed architectures leverage edge AI and sensor fusion to support synchronisation in bandwidth-limited scenarios, which is also a trending research direction [38].

To summarise, a digital twin is the convergence of a virtual model with the real world in real-time, aimed at improving understanding and decision-making. This foundational concept will be referenced throughout our discussion of how digital twins can assist space and remote construction. By having a firm

grasp of what digital twins are and what they can do, we can better evaluate their current use and untapped potential in the following sections.

## 1.2 | Current State and Technological Maturity in Construction and Space Sectors

Digital twin technology has reached markedly different levels of maturity in the space sector (and related aerospace industry) versus the construction sector. Understanding this disparity provides context for how much work remains to fully leverage digital twins in construction, especially for extreme environments.

**Space Sector Maturity:** The aerospace and space exploration community is widely regarded as an early adopter and leader in digital twin development. NASA and major aerospace companies have used high-fidelity simulations and digital replicas for decades as part of design, testing, and operations. In fact, aerospace was *'the first industry to disseminate the concept and use of DTs,'* starting around 2010 when NASA formally integrated digital twins into its technology roadmaps [2–4]. Since then, the sector has produced a number of prominent digital twin applications:

- **Spacecraft and Aircraft Systems:** Companies like Boeing and Airbus employ digital twins to monitor the health of aircraft in operation and improve designs. A well-cited example is Boeing's use of a digital twin for the 787 Dreamliner to analyse its electrical battery systems after service incidents. In the space realm, NASA created digital twins for the James Webb Space Telescope (JWST) during its development. Because JWST's huge folded structure couldn't be fully physically tested on Earth, NASA built a digital twin to simulate its thermal behaviour in orbit and



to rehearse the complex deployment of its sun shield, which had '344 different ways to fail'—in a virtual environment. This use of digital twins was instrumental in ensuring JWST's successful commissioning. Space launch systems similarly benefit: NASA's Space Launch System (SLS) rocket and the Orion spacecraft for Artemis have digital models used to predict performance and guide design certification for parts that cannot be easily tested at full scale.

- **On-Orbit Operations and Satellites:** Satellite operators use digital twins to manage constellations and predict orbital decay or collision risks. As an illustration, SpaceX utilises digital twin technology to optimise the design and testing of its Starship, simulating the entire lifecycle of the vehicle under extreme conditions to identify issues before actual flights. The Starship programme's iterative rapid prototyping approach is underpinned by intensive simulation (essentially treating each prototype and launch as feeding data back into a refined digital model). Even for astronaut training, mixed-reality simulations incorporate digital twin elements: one training platform, drawing on NASA and ESA systems, virtually replicates the ISS (International Space Station) and SpaceX's Falcon Heavy rocket so that astronauts can practice missions and maintenance tasks in a realistic digital environment.
- **Planetary Rovers and Landers:** NASA's Mars rover teams have long used a sort of digital twin for operations. For example, the Curiosity and Perseverance rovers on Mars each have an Earth-based replica and a physics-based simulator that is fed telemetry from Mars, enabling operators to try out manoeuvres virtually before commanding the real rover. This reduces risk when driving over unknown terrain or using the robotic arm. The concept is now extending to the Moon: as part of the Artemis programme, NASA's Lunar Surface Innovation Initiative has developed a digital twin of a proposed regolith excavator robot (the ISRU Pilot Excavator, IPEX). Student teams in a recent NASA challenge had to write software agents to navigate this digital twin of the lunar excavator in a simulated moon terrain, demonstrating how future lunar robots will likely be managed via their digital counterparts.

Overall, the space sector has achieved a high technological readiness for digital twins in specific domains (design/testing of vehicles, spacecraft operations). Many space digital twins are at Technology Readiness Level (TRL) 7–9 (demonstrated in the operational environment or actual use) for specific applications such as spacecraft health management. Agencies such as NASA and ESA continue to invest heavily: NASA's 2022 Digital Twin Strategic Framework explicitly identifies digital twins as a key 'technology foundation' for enabling complex missions and autonomy. There are, however, still challenges specific to space. For instance, dealing with limited bandwidth and intermittent connectivity (a Mars habitat twin must sync data over a 20-min one-way delay), and ensuring models remain accurate under the extreme physics of space (microgravity, radiation effects, etc.). But directionally, the trend is clear: as missions become more autonomous and farther from Earth, digital twins become *indispensable* for mission control and for the assets themselves to manage uncertainty.

**Construction Sector Maturity:** By contrast, the architecture, engineering and construction (AEC) industry has been slower in adopting digital twins and is considered to be in the early stages of maturity. The concept of digital twins entered construction discussions more recently, piggybacking on the spread of BIM and smart infrastructure initiatives [1, 16, 17]. Several factors contribute to this lag:

- The construction industry is traditionally fragmented and project-based, which makes standardisation of digital twin approaches difficult. Early adopters exist, but there is not yet a widely adopted standard or framework for what a construction digital twin should include, unlike aerospace, where NASA and others set clear digital twin definitions early on. The terminology can be inconsistent (some firms label enhanced 3D models as 'digital twins' even if they lack live data links).
- Most current applications in construction are limited in scope. Digital twins in building operations (sometimes called 'smart building' twins or city twins) are being used to monitor energy usage, HVAC performance, or space utilisation in real-time. For example, a commercial building might have a digital twin that integrates its BIM model with IoT sensors for temperature and occupancy, helping facilities managers optimise HVAC settings to save energy. Similarly, infrastructure operators (such as bridge or rail owners) have begun to create digital twins of assets to track structural health. For example, the UK has piloted digital twins for bridges where sensor data on strain and vibration feeds into a model that predicts maintenance needs.
- There are notable pilot implementations. Singapore's 'Virtual Singapore' is a digital twin of the entire city to aid urban planning. Various large construction/engineering firms have been experimenting; for instance, Kiewit and Fluor have used digital models linked with construction site sensors (drones doing reality capture to update progress in the model). In one case, a Chinese project created a digital twin of a new high-rise during construction to track deviations and prevent schedule delays, essentially a 4D (3D + time) twin that updated daily with actual site data. These efforts show promise but are not yet routine on all projects.
- Industry analyses show rapidly growing interest. The global market for digital twins in construction is expanding at ~30%+ CAGR, projected to reach tens of billions USD by the end of the decade. This growth is fuelled by the intersection of BIM, IoT, and project management tools. However, moving from interest to full deployment is slow due to cost and skill barriers.

Despite being behind aerospace, the trajectory is positive. Many construction firms now recognise digital twins as key to digital transformation in construction (often termed 'Construction 4.0'). A recent survey indicated a sharp rise (~40x increase from 2010s to early 2020s) in research publications linking 'digital twin' with construction and maintenance topics, reflecting burgeoning R&D. Industry consortia (such as the Digital Twin Consortium, which includes some AEC companies) and standards bodies are working on reference architectures specifically for construction and infrastructure digital twins.

In summary, the space sector is a decade or more ahead in applying digital twins operationally, whereas the construction sector is catching up rapidly but from a lower base of maturity. The construction industry's current use of digital twins is often in isolated areas (such as building management or construction logistics), and there remains a lack of full integration across an asset's lifecycle and across different systems. However, for the purposes of this article, which deals with *remote and extreme environment construction*, it is worth noting that these kinds of projects (space missions, Antarctic bases, etc.) tend to be one-off, high-stakes endeavours more akin to aerospace projects than conventional construction. Thus, they may benefit from adopting the aerospace mindset and maturity in digital twin usage. Indeed, many remote construction projects are run by governments or large organisations (e.g., a space agency or a mining corporation) that can marshal the necessary high-tech approach. As we move into specific applications in the following sections, we will see examples where construction in extreme environments is leveraging aerospace-like digital twin techniques, effectively narrowing the maturity gap in those niche scenarios.

## 2 | Applications of Digital Twins in Space Construction

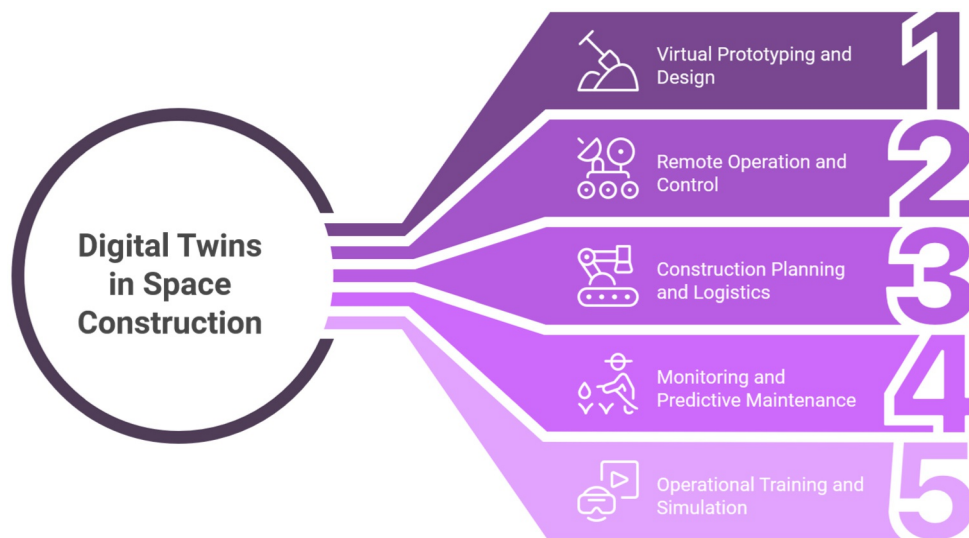
Space construction refers to the building of structures and infrastructure beyond Earth, including on the lunar or Martian surface, in orbit (orbital platforms, space stations, large telescopes), or even in transit (spacecraft assembly). These activities are inherently remote (far from human operators) and occur in harsh conditions that preclude the constant presence of workers. Digital twins are poised to play a pivotal role in all phases of space construction by enabling comprehensive virtual prototyping, remote operations, and predictive maintenance capabilities. Below, we explore key application areas and provide examples (Figure 3):

### 2.1 | Virtual Prototyping and Design of Space Habitats and Structures

Before committing to physical construction on the Moon or Mars, engineers can build a complete digital twin of the proposed structure or habitat. This goes beyond static CAD models; the twin can incorporate local environmental conditions (e.g., lunar gravity, radiation, regolith properties) and simulate the construction process. For instance, NASA and its partners have developed detailed simulations of a lunar base to test building methods such as 3D printing with regolith. A Chinese research team at HUST's National Digital Construction Technology Center created a 'laboratory sandbox' that simulates a future lunar base, where they experimented with various habitat designs (dome structures, egg-shaped modules) under virtual lunar conditions. This digital sandbox allowed them to evaluate how different shapes would withstand the Moon's low gravity, temperature extremes (~300°C swings), moonquakes, and radiation, leading them to an optimised 'lunar pot' double-dome concept. Such virtual prototyping, essentially a digital twin of a not-yet-built habitat, is invaluable. It lets architects identify optimal designs that balance structural strength, material use, and habitability without costly real experiments on the Moon. Moreover, using physics-based twins, one can simulate assembly methods, for example, programme a swarm of construction robots in the digital environment to see if they can autonomously build the structure piece by piece (this ties into swarm robotics, discussed later) [39–41].

### 2.2 | Remote Operation and Control of Construction Robots

On-site human labour in space will be minimal. Construction on the lunar surface or Mars will heavily involve robotic systems, rovers, excavators, manipulators, and 3D printers, which may be tele-operated or semi-autonomous. Digital twins enable



**FIGURE 3** | Application of digital twins in space construction. (Developed by the author for this paper).

a paradigm where these robots are operated through their virtual avatars. Instead of directly joystick-driving a rover with a time delay, an operator (or AI agent) interacts with the rover's digital twin in a simulated real-time environment, and commands are then executed by the physical rover, with the twin providing continuous feedback. Japan's construction machinery company Komatsu has explicitly endorsed this approach for lunar construction: In a government-sponsored programme for autonomous lunar construction, Komatsu is developing digital twin technology to precisely recreate lunar site conditions and machine behaviour in a virtual environment. Their goal is that an excavator on the Moon can be controlled 'without being anywhere near it' by using the digital twin as the control interface. The digital twin would show, for example, the excavator's exact position on the Moon, the terrain elevation map, and the status of its systems, allowing operators on Earth to drive it as if playing a realistic video game—thus overcoming the distance and communication lag. This concept was proven on Earth in analogous scenarios: Hitachi in 2024 reported a 3D digital twin system for remote construction machines on Earth, where sensor-laden bulldozers and cranes were controlled from afar with an interactive twin providing an 'intuitive sense' of location and movement. The result was a safer and easier remote operation that eliminated the need for expert skills, since the operator could essentially see a live virtual replica of the site and equipment. Translating this to the Moon, digital twins can allow relatively unskilled operators (or AI autopilots) to coordinate multiple robots digging regolith, assembling habitat modules, laying cables, etc., all via a synchronised virtual site model. NASA's use of the IPEX excavator's twin in student competitions hints at this future, the students' autonomous software had to navigate the digital twin of the robot, essentially treating it as the real thing [42–44].

### 2.3 | Construction Planning, Logistics, and Simulation

Space construction projects require meticulously choreographed logistics; every material is precious, and every error is magnified by distance. Digital twins of the construction process can simulate the entire workflow. For example, consider building a lunar landing pad from lunar concrete: a digital twin could simulate bulldozers collecting regolith, a regolith processing plant making bricks, robots laying the bricks, and so forth. Planners can vary parameters in the twin (number of robots, task sequences, energy availability, etc.) to optimise the plan prior to execution. This was illustrated by the European Space Agency (ESA) in analogue tests for a lunar base construction: ESA created virtual construction sites to test how autonomous robots might work together to build a lunar habitat, identifying bottlenecks such as one robot waiting on another and then tweaking the plan digitally. In one scenario, a swarm of small robots was simulated to coordinate similar to ants (inspired by ant colony behaviours) to excavate regolith and build a berm for radiation shielding. The digital twin environment allowed the testing of different swarm algorithms and communication strategies without risking hardware. In NASA's case, they have considered the in-space assembly of large structures (like assembling a space telescope in orbit using robotic arms); here again, a digital twin of the structure and robots can run through

the entire assembly sequence many times to iron out issues, much like how aerospace uses 'digital rehearsals' for complex deployments (as with JWST's sunshield deployment twin) [45–47].

### 2.4 | Monitoring and Predictive Maintenance of Space Infrastructure

Once a structure or system is built in space, its digital twin becomes the central tool for monitoring health and scheduling maintenance. Space habitats or vehicles endure harsh conditions (thermal cycling, radiation, micrometeoroids) and must often operate autonomously for long periods. A digital twin can continuously ingest data from structural sensors (strain gauges, temperature sensors, leak detectors, etc.) on, say, a lunar habitat and compare those readings against expected values from the twin's models. If the pressure in a section is dropping or a vibration signature changes, the twin (via algorithms) might detect a probable leak or fatigue crack and alert operators before a catastrophic failure. This approach is akin to how modern jet engines on Earth are monitored by digital twins for early signs of trouble. On the ISS, a rudimentary form of this exists where ground control runs simulations using live telemetry to predict when a system will break and needs replacing. Digital twins would enhance that by being higher fidelity and possibly on-site (running on the habitat's computer for immediate responses). An example in development is NASA's Gateway (the planned lunar orbit station); engineers are creating detailed digital models of Gateway's modules with the intention of using them for operations and troubleshooting once Gateway is in orbit, effectively having a virtual version of the station on the ground that mirrors the real one in space. Another example is the resilience of large space structures: as mentioned, JWST had a digital twin to monitor its mirror temps; future mega-structures (such as a Mars transit ship or a lunar base's power system) will likely have persistent twins for continuous diagnostics. The predictive maintenance benefit is crucial when sending repair parts or crews is expensive and delayed—the twin can forecast when a component (e.g., a solar panel or life support filter) will degrade so that spares and repair missions can be planned well in advance [3, 15].

### 2.5 | Operational Training and Simulation (XR for Construction Tasks)

Digital twins can be combined with extended reality (XR), including virtual reality (VR) and augmented reality (AR) to train astronauts and robotic operators for construction tasks. Before attempting a construction task on the Moon, the crew or robotics team on Earth can practice it in VR using the digital twin model of the site and the tools. NASA has embraced this concept: one training simulation uses digital twins of spacecraft subsystems and realistic physics in an immersive VR setup to train astronauts on launch procedures and EVAs (spacewalks). For construction, imagine an astronaut training to deploy a fold-out habitat or to operate a drilling rig on Mars. They would wear a VR headset that places them in a high-fidelity digital twin of the environment, complete with correct terrain and equipment



behaviour, enabling safe and practical training. When they later perform the real task on Mars, their suits' AR displays could even show live guidance overlayed from the digital twin (for instance, highlighting where the next connector goes on a structure, based on the twin's design). ESA has likewise looked into using AR with digital twins for guiding astronauts in assembling structures, as part of future lunar surface missions. This combination of twin + XR makes the remote construction process much more intuitive and robust [48–50].

In summary, digital twins serve as the nerve centre for space construction activities. They allow engineers to virtually be on the Moon or Mars, designing, simulating, monitoring, and controlling as if they were local, when in reality, they might be hundreds of thousands of kilometres away. By doing so, digital twins dramatically increase the feasibility of ambitious projects such as building a Moon base. As Komatsu's team succinctly put it, '*digital twin technology is the key to outer space construction*' because it enables precise recreation of site conditions and machine behaviour in the digital realm. The next section will see that many of these same capabilities, remote monitoring, simulation, and predictive analytics, are equally game-changing for construction in Earth's remotest corners.

### 3 | Applications of Digital Twins in Remote Terrestrial Construction

Remote terrestrial construction refers to building or managing infrastructure in locations on Earth that are extremely difficult to access regularly by people. This includes polar regions (Arctic and Antarctic sites), undersea environments, deserts and other isolated landscapes, and high-risk zones. Such projects face some analogous challenges to space construction: limited on-site personnel, harsh weather or environmental conditions,

logistical complications, and high stakes for failure. Digital twins can be leveraged in these contexts to improve planning, enable remote supervision, and ensure resilience (Figure 4).

We will discuss a few representative environments and how digital twins are applied:

#### 3.1 | Arctic and Antarctic Infrastructure

The far north and south of our planet host critical infrastructure such as research stations, roads, pipelines, and mines, all under severe cold and inaccessibility. Climate change is further destabilising these environments (e.g., thawing permafrost undermining foundations). Digital twins are emerging as vital tools to monitor these assets in real-time and predict issues caused by environmental changes. For example, in Northern Canada and Alaska, many buildings and pipelines are built on permafrost. Engineers now use digital twins of these structures to track ground temperature, soil movements, and structural responses continuously. One policy report noted that '*among the most promising tools is Digital Twin technology, which offers an evolving picture of how an asset is responding to environmental conditions... for infrastructure built on permafrost, this means engineers can track changes in ground temperature and moisture over time and predict how these changes might impact the stability of a road, pipeline, or building*'. Concretely, this might involve sensors in the ground measuring permafrost thaw depth and in the structure measuring tilt or strain; the twin assimilates these data and runs thermal-structural models to forecast when the foundation will reach a critical level of settlement. Early warnings from the twin can trigger interventions (such as installing ground chillers or shoring up supports) to prevent catastrophic failures due to permafrost collapse.

| Characteristic                                                                                         | Arctic/Antarctic                                        | Underwater                                                | Desert/Land                                           | Long-Distance Infra.                                     | Disaster Response                                         |
|--------------------------------------------------------------------------------------------------------|---------------------------------------------------------|-----------------------------------------------------------|-------------------------------------------------------|----------------------------------------------------------|-----------------------------------------------------------|
|  <b>Monitoring</b>  | Real-time asset monitoring and issue prediction.        | Continuous analysis of sensor data for anomaly detection. | Real-time data collection for maintenance prediction. | Segment-wise monitoring for issue detection.             | Guiding remote-controlled equipment for debris clearance. |
|  <b>Maintenance</b> | Optimal timing and resource planning for repairs.       | Reduced physical inspections through virtual analysis.    | AI-guided dispatch of cleaning robots or crews.       | Predicting frost heave and scheduling re-stabilization.  | Planning robotic construction of containment walls.       |
|  <b>Operation</b>   | Efficient ship operation through fuel usage simulation. | Remote operation of unattended facilities.                | Remote operations and performance optimization.       | Simulating oil flow and temperature profiles.            | Remote operation of machinery to minimize human exposure. |
|  <b>Assessment</b>  | N/A                                                     | Environmental impact assessment via simulation.           | Simulating different mine plans for efficiency.       | Cross-referencing geospatial and engineering data.       | N/A                                                       |
|  <b>Benefits</b>    | Improved planning, remote supervision, and resilience.  | Enhanced maintenance and monitoring of subsea assets.     | Optimized production and problem detection.           | Early detection of issues like leaks or ground movement. | Minimizing human exposure in hazardous zones.             |

**FIGURE 4** | Applications of digital twin technologies across extreme and critical environments, highlighting their roles in monitoring, maintenance, operation, assessment, and overall benefits. (Developed by the author for this paper).



Moreover, digital twins help with the maintenance logistics in these regions. Because sending repair crews to the Arctic is expensive and seasonal, a twin can tell you the optimal window to fix something and exactly what is needed. For instance, a digital twin of a remote Arctic bridge could predict that based on current rust progression (tracked via sensors or inspections), a specific joint will need reinforcement within 2 years, allowing time to schedule a summer airlift of materials and engineers, rather than being caught off guard in winter. The British Antarctic Survey (BAS) has used digital twin technology on its research ship RRS Sir David Attenborough to reduce carbon emissions and improve maintenance. They created a twin of the ship's power and propulsion system to simulate fuel usage and optimal settings under various conditions. This helped operators run the ship more efficiently (a sustainability win) and also test out upgrades digitally while the ship was in Antarctica, far from any port [51–53].

### 3.2 | Underwater Construction and Subsea Infrastructure

The ocean environment presents continuous challenges for the construction of oil rigs, subsea pipelines, wind turbines, and underwater tunnels. These assets are submerged, not easily reachable for inspection, and subject to corrosion and pressure stresses. Digital twins have become integral in the subsea industry for maintenance and monitoring. A notable use case is the monitoring of offshore oil and gas infrastructure. Traditionally, companies send divers or ROVs (remotely operated vehicles) every few years to inspect platforms and pipelines. Now, with digital twins, they can drastically reduce physical inspections. For example, the National Subsea Centre in the UK highlighted how a digital twin of a network of subsea pipelines continuously analyzes sensor data (pressure, flow, vibration) to detect anomalies such as small leaks or fatigue cracks. If a sensor detects a slight pressure drop in a pipeline segment, the twin's model might localise it to a potential pinhole leak and schedule a targeted ROV inspection at that spot rather than a blind full survey. The twin can also simulate the pipeline's integrity under different conditions, for example, how it would handle a 50-year storm or varying temperatures to guide preemptive maintenance.

Remote operation is another aspect: many new offshore installations are designed to be normally unattended facilities (NUFs). Their operation is controlled from onshore control centres, sometimes hundreds of kilometres away. Digital twins serve as the virtual interface for operators to manage these remote platforms. The twin of a subsea well can show valve states, flow rates, and even a 3D visualisation of the equipment on the seafloor, allowing engineers to open/close valves or adjust pumps remotely with confidence, as if they had an eye on the seabed. This mirrors what has been done in mining and space: operate via the twin to compensate for not physically being there.

Environmental impact assessment is also aided by digital twins underwater. As noted by NSC, a twin can incorporate data on water quality, currents, and marine life around a subsea structure. By simulating potential incidents (such as an oil leak or noise from construction) in the twin, companies and regulators can predict the impact on the ecosystem and take measures to

mitigate harm. This proactive use of twins aligns with sustainability goals (e.g., planning construction timing to avoid marine mating seasons by simulating noise propagation in the ocean, something a twin can do by coupling acoustic models with real hydrophone data) [54–56].

### 3.3 | Desert and Remote Land Sites

Deserts such as the Sahara, Australian Outback, or Gobi are often chosen for massive projects such as solar farms, mines, or pipelines. These areas have extreme heat and dust and are far from population centres, making on-site staffing difficult. In such projects, digital twins support primarily remote operations and performance optimisation. A solar farm in a desert might have a digital twin that collects real-time data from each panel (temperature, output) and from weather stations (sunlight, dust levels) to predict maintenance needs such as panel cleaning or cooling requirements. With potentially thousands of panels, the twin can use AI to identify which sections are underperforming due to dust accumulation and then guide autonomous cleaning robots or dispatch a small crew directly to those spots [57–59].

Mining operations in remote deserts have been early adopters of digital twin-like approaches. Companies in Australia, such as Rio Tinto, operate large iron ore mines in the Pilbara desert from control centres in Perth (over 1500 km away). They use integrated digital models of the mine sites, fed by data from autonomous haul trucks, drilling rigs, and railway sensors. Essentially, these are digital twins of the mines, complete with topography updated by drone surveys and IoT data from machinery, allowing managers to optimise production and detect problems without setting foot in the mine pit. One report mentions that top mining firms are 'pioneering Digital Twin and AI strategies to enhance productivity and safety' in these remote mines. The benefits include being able to simulate different mine plans (such as digging sequences) in the twin to see which yields the best efficiency or to foresee bottlenecks in the system (such as a breakdown of a critical truck). Additionally, by linking the twin to AI, anomalies such as an overheating truck engine can be caught via patterns in the data, and the truck can be rerouted to maintenance before it fails in the field. This predictive maintenance via twin is directly credited with reducing downtime in these remote mines.

A specific example: Rio Tinto's Gudai-Darri mine (formerly Koodaideri) is branded as the company's most technologically advanced mine. It uses a 'digital twin of the mine' to integrate data from all assets and systems, a concept that Rio Tinto acknowledges was inspired by NASA's pioneering use of twins. This twin coordinates autonomous drills, trucks, and even a robotic train system (AutoHaul). The outcome is that personnel can monitor and control the entire mining operation from a single ops centre building, with the twin providing situational awareness as good as being on-site.

### 3.4 | Monitoring Long-Distance Infrastructure

Infrastructure that stretches over remote areas, such as long pipelines, electricity transmission lines, or highways through

uninhabited terrain can benefit from digital twins for segment-wise monitoring. For pipelines, which often traverse deserts or tundra, companies are implementing digital twin systems that take in flow data, pressure readings, and even satellite imagery to detect issues such as leaks or ground movement affecting the pipeline. If a landslide in a desert canyon exposed a pipeline, normally, it might go unnoticed for weeks; a well-designed twin, however, could combine remote sensing data with pipeline strain gauge data to flag the anomaly within hours and pinpoint the location for drone inspection. This cross-referencing of geospatial data and engineering data is a strength of digital twins, creating what is essentially an ‘asset intelligence network’ over huge distances. The operators of the Trans-Alaska Pipeline, for instance, have explored using a digital twin to simulate oil flow and temperature profiles along the 800-mile pipeline in conjunction with permafrost data, so they can predict where frost heave might cause the pipeline to shift and schedule preventive re-stabilisation [60–62].

### 3.5 | Extreme Disaster Response Construction

Another remote scenario is when a disaster (earthquake, hurricane) wipes out infrastructure, and rapid rebuilding is needed in a hazardous zone. Teams cannot always be on site (due to aftershocks or contamination). Digital twins of the affected area, constructed from satellite and drone data, can guide remote-controlled construction equipment to clear debris or erect temporary structures without putting responders in harm's way. Although not routine yet, this application has been theorised: for example, after a nuclear accident (such as Fukushima), a digital twin of the site with radiological maps could be used to plan robotic construction of containment walls and remotely operate the machinery, minimising human exposure.

Across these examples, the recurring theme is that digital twins enable a shift from reactive site-bound operations to proactive remote and data-driven operations. By having a continuously updated virtual replica, organisations can make informed decisions without having eyes physically on the asset 24/7. This is transformative for remote construction, where travel to the site might be infrequent or dangerous. Furthermore, digital twins contribute to sustainability and efficiency in these projects: maintenance can be done just in time (reducing unnecessary trips or component replacements), operations can be optimised for energy use, and the lifespan of assets can be extended by carefully monitoring and mitigating deterioration. For instance, if a twin shows that reducing a remote diesel generator's load at certain times will extend its life by a year, operators can adjust accordingly and avoid having to fly in a replacement so soon, saving cost and reducing carbon footprint [63–65].

It is worth noting that while the technology is promising, implementing digital twins in these environments requires overcoming challenges such as sparse connectivity (e.g., Antarctic stations have limited bandwidth to send data to their twins in real-time), sensor reliability in harsh conditions and cybersecurity (remote control via a twin must be secure against hacking). Nonetheless, successful case studies are accumulating, as we will highlight in the next section.

Beyond the specific contexts of Arctic, subsea, desert, and disaster zones, digital twins also offer transformative benefits for other remote terrestrial infrastructure systems facing operational and environmental challenges. For example, digital twins are increasingly used in urban critical infrastructure networks, such as water distribution systems, power grids, and transportation networks, to improve resilience and continuity under climate-induced risks. Urban Digital Twin (UDT) systems integrate real-time sensor data, predictive analytics, and simulation models to support decision-making under uncertainty, optimise resource allocation, and enhance service delivery across extended networks vulnerable to floods, heat waves, or supply disruptions [66]. Studies show that digital twins can significantly improve resilience and sustainability in critical infrastructure such as utility networks by enabling real-time anomaly detection and scenario planning, which reduces downtime and service interruptions [36]. Moreover, infrastructure digital twins in smart cities are being used to forecast social and environmental stresses (e.g., anticipating load peaks or infrastructure failures), demonstrating that remote terrestrial use cases are not limited to isolated sites but extend to networked urban-regional systems that are complex, geographically dispersed, and subject to systemic risks [67].

## 4 | Global Case Studies

To illustrate how digital twin technology is being applied in both space and remote terrestrial construction, we present a range of case studies from around the world involving government agencies, space organisations, and industry. These examples demonstrate the real-world impact and lessons learnt from deploying digital twins in extreme environment projects:

### 4.1 | NASA—Artemis Programme and Lunar Infrastructure

NASA has integrated digital twin concepts throughout the Artemis lunar exploration programme. *Case in point:* Artemis Lunar Base Planning, NASA's analogue studies for a lunar base (such as NASA Langley's habitat testbeds) rely on digital twins to iterate designs. They created high-fidelity digital models of habitats and rovers, incorporating lunar environmental data, to decide optimal layouts for Artemis Base Camp. Additionally, NASA's Lunar VIPER rover (set to prospect lunar ice) has a full digital twin for mission simulation; engineers at Ames Research Center use it to simulate driving routes and drilling operations in lunar soil before the rover goes to the Moon, reducing risk. Strategically, NASA sees digital twins as crucial for sustaining long-term lunar operations where Earth contact is limited. Policy-wise, NASA's leadership has been evangelising digital twins; for example, in 2021, NASA released a comprehensive plan (NASA Technical Memo 20220018538) outlining how digital twins, combined with AI and IoT, will enable ‘faster, more agile processes’ and safer missions [68].

An illustrative deep-dive into NASA's use of digital twins is found in its critical infrastructure and aerospace operations. In the context of space missions and critical infrastructure, digital

twins have been shown to significantly advance real-time decision support and predictive resilience. Ref. [36] describe how integrated digital twin frameworks improve uptime, security, and efficiency across power, transportation, and critical utilities by bridging physical systems with virtual analytics. NASA's Artemis programme extends this principle to space habitats and vehicles: digital twins are constructed to ingest sensor data, simulate environmental stresses, and forecast system degradation before failures occur, enabling mission planners to prioritise maintenance and risk mitigation under significant communication delays. This is directly analogous to critical infrastructure twins on Earth that integrate IoT streams and predictive models to optimise maintenance cycles and reduce unscheduled outages. Studies also point to the broader applicability of such frameworks in urban and smart infrastructure resilience, highlighting that the design, testing, and validation processes pioneered by NASA influence network-scale twin ecosystems used for complex terrestrial infrastructure.

#### 4.2 | European Space Agency (ESA)—Analogue Testing and 'Digital Twin Earth'

ESA's contributions span both space construction and Earth system twins. On the space side, ESA has run analogue field trials (e.g., Pangaea and Luna simulations) where astronauts and robots perform mock lunar construction tasks; these trials use digital twins of equipment and sites to evaluate operations. On Earth, ESA is spearheading the 'Digital Twin Earth' project under the EU's Destination Earth initiative. Although not about construction, Digital Twin Earth is a platform simulating Earth's climate and natural systems at high fidelity to inform climate resilience strategies, a concept directly relevant to designing resilient infrastructure. Insights from Digital Twin Earth (e.g., predicting sea-level rise or extreme weather patterns) feed into how we should build future coastal or Arctic infrastructure. This demonstrates cross-pollination: space agencies are applying their digital twin expertise gained from spacecraft to broader planetary management and infrastructure policy, effectively treating Earth itself as a 'remote environment' that needs a twin for better stewardship [69].

#### 4.3 | Japan (JAXA) and Komatsu—Autonomous Lunar Construction Project

As mentioned earlier, Japan's Ministry of Land, Infrastructure, Transport, and Tourism (MLIT) initiated a multi-company project for Autonomous Construction in Space (the 'Stardust Programme'). Komatsu, a leading construction machinery manufacturer, was selected for its proposal to use digital twins on the Moon. In practice, Komatsu is developing a suite of construction equipment (bulldozers, excavators) that can operate on the lunar surface. The digital twin component involves creating a virtual lunar worksite with realistic physics. Komatsu has been testing controlling earth-moving robots in a simulated lunar terrain (including latency to mimic the Moon-Earth delay) via their digital twins, and initial results showed operators could grade soil remotely with only virtual feedback. This case study is significant because it is a public-private partnership: JAXA

(Japan's space agency) provides lunar data and requirements, MLIT provides funding and integration to national space strategy, and Komatsu provides technology and machinery know-how. It shows how policy and industry can align; MLIT sees value not just for space but also in how the tech can rebound to benefit remote construction in Japan (e.g., operating construction machines in disaster zones or hazardous areas via twins). This programme thus has dual-use: advancing lunar base capabilities and modernising domestic construction automation [42, 43].

#### 4.4 | SpaceX—Starship Development and Mars Mission Simulation

SpaceX, a private company, has aggressively used digital simulations which effectively function as digital twins. The Starship vehicle (intended for Moon and Mars missions) is so large and novel that SpaceX cannot test everything physically. Instead, per reports, SpaceX created a digital twin of Starship to simulate its entire flight envelope and even a hypothetical Mars mission. A 2024 scientific paper noted that SpaceX's team runs simulations 'based on publicly released data' to validate Starship's design for a human Mars mission, including interplanetary trajectory and entry, descent, and landing sequences. In one instance, SpaceX's twin revealed structural issues in ground tests, prompting design tweaks before the next prototype was built, saving time and money (this is partially evidenced by the relatively quick iterations of Starship prototypes). Another facet: SpaceX's launch facilities at Boca Chica, Texas, are heavily instrumented and modelled; after the first Starship orbital launch attempt caused pad damage, SpaceX used a digital twin of the pad and rocket plume to redesign a new water-cooled flame diverter for the launch mount. By simulating the next launch with this twin, they improved the pad's resilience. SpaceX's case underlines the private sector's willingness to rely on digital twins for rapid innovation. It also hints at how a Mars base by SpaceX might be managed. Elon Musk has mentioned having 'ops centres on Earth controlling things on Mars', which implies extensive digital twin infrastructure to supervise those distant operations [70].

#### 4.5 | ispace and Other NewSpace Companies

ispace, a Japanese startup, attempted a lunar landing in 2023 (Mission 1). Although not much is publicly disclosed about their use of digital twins, it is known that they ran extensive simulations of the lander's descent and had a mission control that visualised the lander's state in real-time. We can infer that a digital twin of the lander (with its dynamics) was used to practice the landing sequence and adjust parameters. Although the lander ultimately crashed, the company stated they would analyse the data via their simulation models to pinpoint what went wrong, essentially using the twin for *post-incident forensic analysis*, similar to how Apollo 13's twin was used. This pattern is common: when new space companies (such as ispace, Astrobotic, etc.) operate at lower budgets, digital twins offer a cost-effective means to test and troubleshoot without needing multiple expensive physical prototypes [71].

## 4.6 | United Kingdom—National Digital Twin Programme

The UK has been a thought leader in applying digital twins for national infrastructure. Through the Centre for Digital Built Britain (CDBB) and initiatives such as the Gemini Principles, the UK government is pushing for ‘an ecosystem of connected digital twins’ to improve infrastructure resilience and services. Although this is more policy and strategy than a single project, it has led to several pilot case studies: one involves a digital twin of London’s water supply network to predict leaks and optimise water pressure (led by utility companies), another is a digital twin of an offshore wind farm to maximise energy output and schedule maintenance during favourable weather windows. For remote construction relevance, the UK’s effort demonstrates how multiple twins can link. For example, a twin of an energy grid can connect with a twin of a remote wind farm and a twin of weather systems to form a holistic view. Policy-wise, the UK’s approach addresses data sharing and interoperability, which is crucial if, say, a space habitat twin should interface with Earth-based systems or if a country wants to combine climate twins with infrastructure twins to plan protective construction [72].

An illustrative case in remote and connected infrastructure is the development of Urban Digital Twins (UDTs) for resilient utility systems. Ref. [66] shows that UDTs integrate real-time sensor feeds, advanced simulations, and predictive analytics to support decision-making across water, energy, and transportation networks under climate stress. For example, digital twins of urban water distribution networks facilitate instant monitoring of hydraulic behaviour and enable intelligent response strategies to pipe failures or demand surges before service disruption occurs, enhancing reliability in the face of extreme events. Similarly, digital twin frameworks for smart grids allow utilities to simulate fault propagation and optimise load balancing under variable environmental conditions, which is critical for networks covering remote and high-latitude regions [67]. These detailed cases demonstrate that digital twin implementations can transition from asset-centric systems to network-wide resilience platforms, offering continuous adaptive capacity for distributed critical infrastructure essential to communities under stress.

## 4.7 | Canada—Mining and Northern Infrastructure

Canadian companies and researchers have deployed digital twins in mining (similar to Australia) and are exploring twins for northern roads. *Case:* Diavik Diamond Mine (Northwest Territories): This Arctic mine uses a digital twin to integrate real-time data from its open-pit and underground operations with its ore processing plant. The twin helps coordinate equipment and predict maintenance, which is vital as the mine is accessible by an ice road only part of the year. By knowing exactly which truck or drill needs servicing and when, Diavik can ensure parts are brought in on the annual ice road or flights, avoiding emergency fixes in winter. Another Canadian project is the SMARTICE programme, which uses a digital twin of sea-ice conditions along Arctic community ice roads (these are seasonal roads over frozen sea used to truck supplies). The twin assimilates satellite imagery and sensor measurements to map ice

thickness and strength, predicting when and where the ice road will become unsafe. This directly informs the construction of alternate land routes or bridges to maintain supply lines as climate warms, showcasing a fusion of indigenous knowledge, IoT, and twin technology for resilience [73].

## 4.8 | Industry—Shell’s Digital Twin for Offshore Platforms

Royal Dutch Shell developed a comprehensive digital twin of its massive Prelude floating LNG platform (offshore Australia), the largest floating structure ever built. This twin includes a 3D model of the entire facility and real-time data from tens of thousands of sensors. During operations, Shell’s engineers (often onshore) navigate the twin in VR to inspect equipment ‘virtually’ instead of doing risky on-platform inspections. In one instance, when a compressor fault occurred, engineers used the twin’s historical data and simulation capability to identify the cause and fix it with minimal downtime, saving millions in lost production. Prelude’s twin is an example of scale: it is similar to a mini-city twin on the ocean. Shell has reported that lessons from Prelude’s twin allow them to design future platforms with better placement of sensors and design for maintainability [74].

These case studies collectively underline several points. First, digital twin adoption is inherently interdisciplinary and often requires collaborations (space agencies with manufacturers, governments with tech firms). Second, the value of digital twins has been proven in both prevention and response, preventing failures (JWST, pipelines, wind farms) and responding to anomalies (Apollo 13, ispace crash analysis, oil rig faults) more effectively. Third, policy frameworks such as the UK’s or Japan’s indicate that governments are recognising digital twins as strategic infrastructure in their own right, deserving investment akin to physical infrastructure because of the insights they provide.

For policymakers, these examples provide evidence that supporting digital twin development (through funding, open data policies, standard-setting) can yield safer, more resilient, and efficient outcomes in construction projects. For industry leaders, the cases demonstrate competitive advantages, companies that leverage twins (SpaceX, Rio Tinto, Shell) can move faster, safer, and often cheaper by anticipating issues and optimising operations.

In extreme environment construction, where uncertainty is high, these advantages become even more critical. A lunar base or a polar railway cannot afford trial-and-error to the same degree as a temperate project; having a digital twin acts as a high-tech insurance policy and guide. The next section will build on this by discussing strategic and policy implications in a broader sense, considering how to maximise the benefits observed in these cases (Figure 5).

## 5 | Strategic and Policy Implications for Future Infrastructure Resilience and Sustainability

The rise of digital twins in construction for extreme environments carries significant strategic and policy implications. As

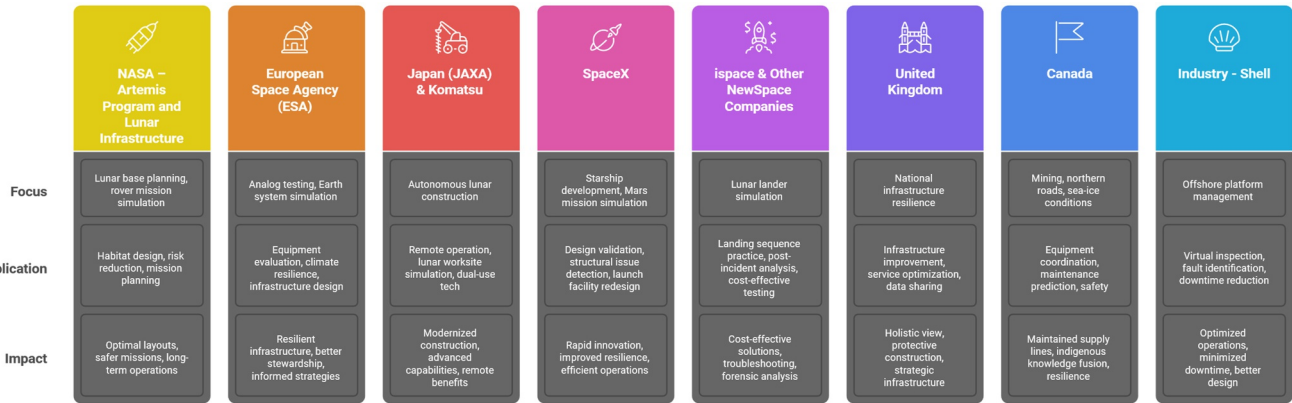


the case studies and applications have shown, digital twins can greatly enhance the resilience of infrastructure and contribute to sustainability, but realising these benefits at scale requires forward-thinking policies, standards, and investments. We highlight key implications and recommendations (Figure 6):

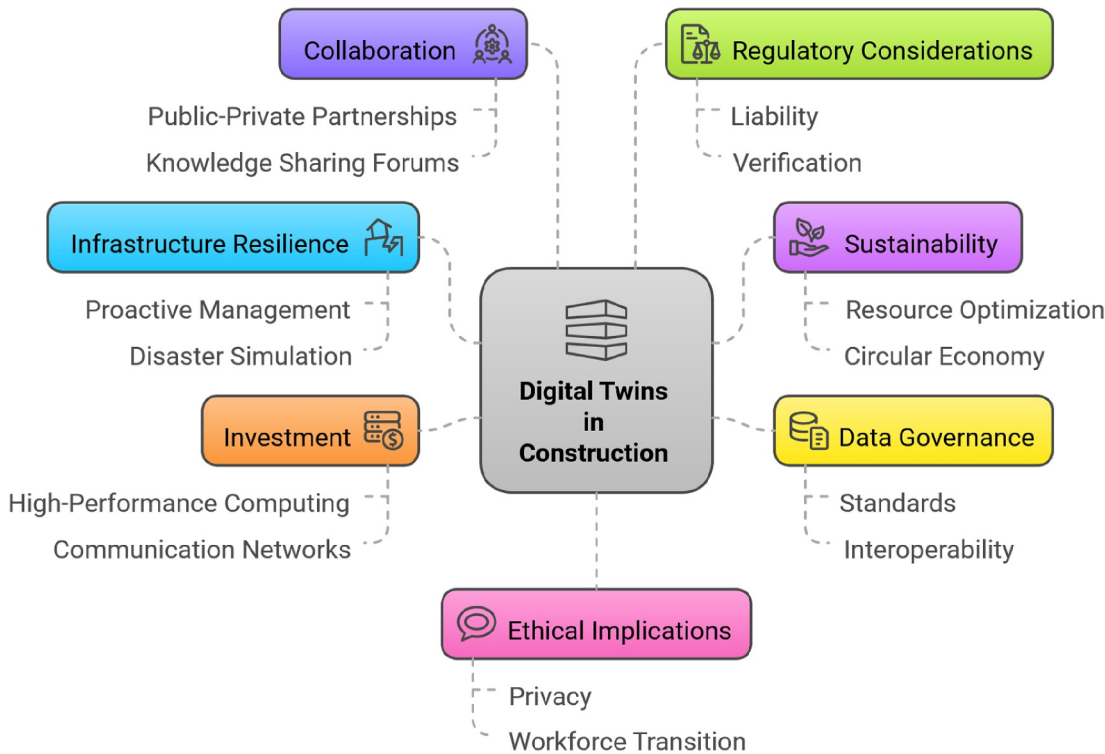
### 5.1 | Infrastructure Resilience Through Proactive Management

One of the most powerful implications is the shift from reactive to proactive infrastructure management. With digital twins, issues can be forecast and mitigated before they escalate, which is a cornerstone of resilience. For policymakers, this means that

encouraging the use of digital twins in public infrastructure (bridges, levees, power grids) could reduce disaster impacts and maintenance costs. For example, a government could mandate that all new critical infrastructure in hazard-prone areas (flood zones, permafrost regions, seismic zones) include a digital twin for continuous monitoring. Over time, this could be written into building codes or concession agreements for large projects. The payoff is huge: less infrastructure failure means fewer emergency repairs and less risk to communities. A twin can simulate disasters (similar to an earthquake's effect on a remote dam) and identify vulnerabilities, guiding where to reinforce ahead of time. In essence, digital twins make infrastructure adaptive, they can feed early warning systems and support real-time emergency decision-making (similar to how NASA's twins



**FIGURE 5** | Global initiatives leveraging digital twin technologies for infrastructure, construction, and mission resilience in space and extreme environments. (Developed by the author for this paper).



**FIGURE 6** | Key enablers and considerations for implementing digital twins in construction, including governance, ethics, resilience, and sustainability. (Developed by the author for this paper).

help in mission anomalies). Governments should consider digital twin data as part of national resilience planning, perhaps integrating these into monitoring centres for natural disasters [75, 76].

## 5.2 | Sustainability and Efficient Resource Utilisation

By optimising operations and maintenance, digital twins contribute to sustainability goals. They help extend the lifespan of assets (through timely maintenance and avoiding catastrophic failures that lead to rebuilds) and reduce resource waste. For instance, twins can optimise energy usage in remote facilities (ensuring generators run at optimal load and coordinating renewable energy sources with demand). They can also minimise unnecessary travel, if you can inspect virtually, you avoid flying a helicopter to an oil rig or driving to a pipeline pump station, cutting carbon emissions. At scale, this means a smaller environmental footprint for maintaining infrastructure. Policymakers focused on decarbonisation can leverage this: they could provide incentives (such as tax breaks or innovation grants) to infrastructure operators who implement digital twins proven to save energy or reduce emissions. Additionally, digital twins can help with circular economy principles. By tracking the condition of materials and components, they facilitate refurbishing or recycling infrastructure parts at end-of-life rather than scrapping them blindly. For example, a twin might reveal that certain modules of a Mars habitat can be safely reused for another mission, informing sustainable mission design [77, 78].

## 5.3 | Data Governance, Standards, and Interoperability

Effective digital twin ecosystems require robust data governance. Twins integrate data across sectors. Consider a remote mining twin that might benefit from satellite weather data or a space habitat twin that might need inputs from an Earth-based medical database for astronaut health. Sharing such data raises issues of privacy, security, and interoperability. Strategically, there is a need for standards to allow different digital twins (maybe owned by various entities) to connect. The UK's Information Management Framework (IMF) for connected digital twins is a leading example, setting out semantic standards and protocols for data exchange. Internationally, organisations such as ISO and IEEE are working on digital twin standards (e.g., ISO 23247 [79] for digital twin in manufacturing). Governments can accelerate this by adopting open standards in their projects and requiring vendors to comply, thus avoiding vendor lock-in and siloed systems. Interoperability is especially crucial in remote scenarios involving multiple stakeholders: for instance, a polar observatory's twin might need to interface with a national climate twin, or a lunar base twin might integrate data from modules built by different countries. A common framework (akin to the internet protocols for data) ensures all these twins can 'talk' to each other in a secure, meaningful way. Trust frameworks (such as the Gemini Principles emphasising trust, security, and public benefit) are also needed so that organisations are willing to share data from their twins when it serves a greater resilience goal [80, 81].

## 5.4 | Investment in Digital Infrastructure

Viewing digital twins as part of infrastructure implies that nations should invest in them similarly to physical infrastructure. This includes funding for high-performance computing, communication networks, and sensor deployments that make twins possible. For example, to have a digital twin of the entire national grid, a country may need to install thousands of extra sensors and a robust fibre-optic network to carry the data. These have costs that can be justified by the benefits of preventing blackouts or enabling more renewable integration via the twin's insights. In remote construction, an investment might be needed for satellite or 5G communications to ensure the twin of, say, a remote dam can update in real-time despite its location. There is also a human capital investment: training engineers and operators to work with digital twins (data analytics skills, familiarity with simulation tools). Policymakers could encourage curricula that blend civil engineering with data science and perhaps create exchange programs between aerospace experts and civil infrastructure experts to cross-pollinate knowledge on twins [75, 82].

## 5.5 | Public-Private Collaboration and Knowledge Sharing

Many digital twin successes have come from one-off projects (e.g., JWST, Rio Tinto's mine), and often that knowledge is not widely disseminated. To avoid reinventing the wheel, there is a need for better knowledge sharing. Governments and industry bodies can organise forums or working groups specifically on digital twins for extreme environments, where lessons learnt can be discussed. NASA, for instance, might share methodologies with mining companies on how they validate twin accuracy, or vice versa (mining might share how they manage twin operations with intermittent comms, which could help space missions). Public-private partnerships can also de-risk the initial development of digital twins for new domains, as Japan did with Komatsu, and as the EU does via Horizon Europe funding for digital twin pilots. By co-funding these initiatives, the public sector ensures the results are openly available to benefit all (e.g., requiring open results or software tools from the projects). One strategic recommendation is establishing a 'Digital Twin Extreme Environments Consortium' globally, involving space agencies, military engineers (who build in extreme places such as Afghanistan mountains or ocean floors), civil engineers, and tech companies, to develop best practices and perhaps open-source toolkits for twins [83, 84].

## 5.6 | Regulatory and Legal Considerations

As digital twins begin to inform and even automate decisions on physical systems, questions arise about liability and regulation. If a digital twin recommends a certain action (e.g., an AI in the twin shuts down a pipeline to prevent damage) and that action causes financial loss or fails to prevent an accident, who is responsible, the operator or the software developer? Regulatory frameworks will need updating to account for AI-driven and digital twin-driven operations. Verification and validation of

twins (especially those used for safety-critical decisions) might need regulatory oversight, similar to how aircraft simulators are certified. We might see requirements for independent audits of important digital twins (for instance, a twin that manages a nuclear power plant or a large dam) to ensure their models are accurate and secure from cyber threats. Policymakers should start to address these issues proactively, possibly by updating infrastructure regulations to mention digital operational systems, and ensuring any automation from twins remains ‘human in the loop’ until clear guidelines are established [85, 86].

## 5.7 | Ethical and Social Implications

With increased monitoring (such as sensors everywhere feeding twins), there could be privacy issues, especially if the infrastructure involves communities (smart city twins, etc.). Even in remote construction, consider that a twin might monitor workers via wearables for safety (common in modern construction). Policies must ensure that the data collected is used ethically and with consent. Additionally, workforce implications: some fear that heavy automation via digital twins could reduce certain jobs (such as on-site inspectors or equipment operators). Policymakers should accompany twin rollouts with retraining programs so that those workers can transition to new roles such as twin supervisors, analysts, or robot operators. These roles are often safer and higher-skilled. The narrative should be straightforward that digital twins augment human decision-making rather than replace it; this framing can help with public acceptance, particularly in government projects [87, 88].

In conclusion, on policy, embracing digital twin technology for space and remote construction aligns with broader objectives of building resilient, sustainable, and smart infrastructure.

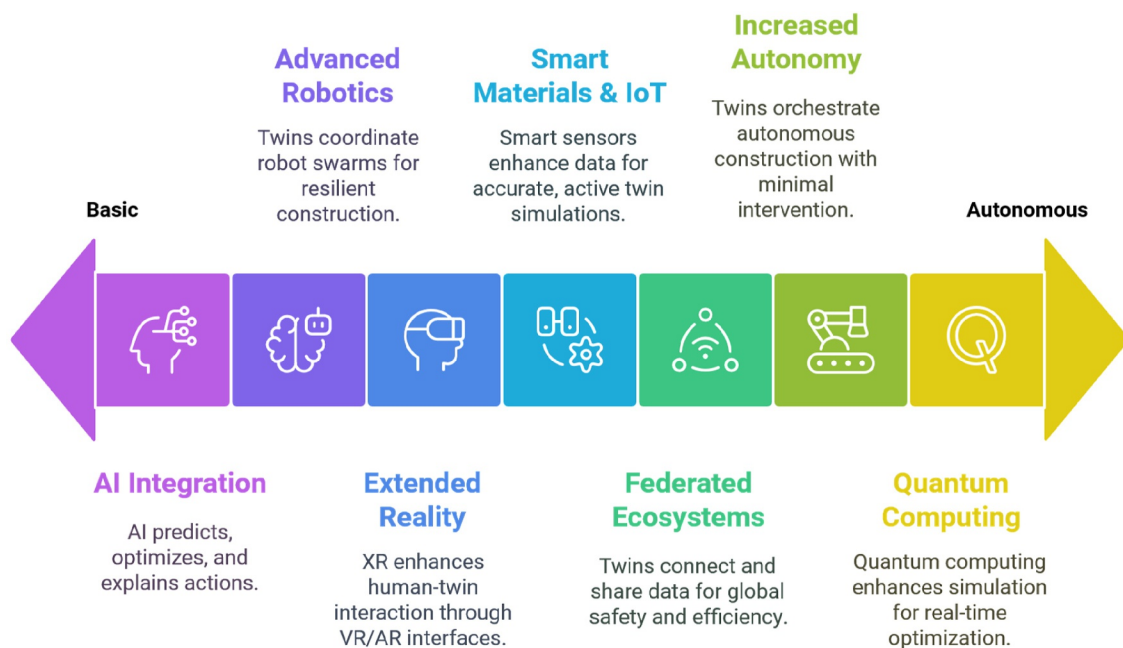
Countries that move early to integrate twins into their infrastructure strategy may gain a significant advantage in managing the impacts of climate change and in carrying out ambitious projects (such as space exploration or Arctic development) safely. The transformative potential of twins justifies treating data and simulations as critical infrastructure themselves. As one report succinctly put it, *connected digital twins are tools to understand the complexities of interconnected systems and provide better insights for decisions*. In our context, those interconnected systems span Earth and space. By making digital twins a strategic priority, policymakers ensure that as we push the frontiers, building cities on the Moon or wind farms at the poles, we do so with eyes wide open, informed by the best virtual foresight technology can offer.

## 6 | Future Trends and Emerging Technologies

Looking ahead, the landscape of digital twins for construction in extreme environments is set to evolve rapidly, propelled by advances in related technologies. Several emerging trends are likely to shape the next 5–10 years, making digital twins even more powerful and indispensable (Figure 7):

### 6.1 | Artificial Intelligence (AI) and Machine Learning Integration

AI is poised to become the ‘brain’ of digital twin systems, handling complex predictions and autonomous decision-making. Currently, many twins use relatively deterministic physics models and rule-based thresholds for alerts. In the future, machine learning algorithms (trained on the vast data that twins accumulate) will identify patterns and optimise operations in ways humans might not notice. For example, an AI



**FIGURE 7** | Emerging technologies driving next-generation digital twin capabilities in construction and infrastructure. (Developed by the author for this paper).

might analyse years of data from a Martian habitat twin and discover subtle signs that precede an airlock failure, then proactively adjust maintenance schedules. AI-driven twins could also run countless simulation scenarios (Monte Carlo simulations) automatically and use reinforcement learning to improve control strategies for robots or processes. NASA indicates that as AI/ML mature and integrate, *'the digital twin will play a more central role in space missions, especially in remote operations and complex system management'*, essentially because AI can help handle non-deterministic, unexpected situations autonomously. We can envision a Mars base twin that has an AI assistant continuously optimising life support settings for efficiency and longevity, or a pipeline twin that uses ML to accurately predict corrosion growth under varying conditions far better than manual models. However, with AI comes the need for transparency. An ongoing area of research will be explainable AI for digital twins, ensuring that when an AI in the loop suggests a critical action (such as 'evacuate habitat Section 3'), it can explain which sensor readings or model inferences led to that suggestion [3, 89].

## 6.2 | Advanced Robotics and Swarm Automation

The future of construction in extreme environments will likely involve robotic teams rather than single machines. Swarm robotics, large numbers of smaller robots working collaboratively, draw inspiration from social insects and promise redundancy and efficiency. Digital twins will be crucial for coordinating and monitoring swarms. Researchers are already exploring how digital twin platforms can handle communication and task allocation among robot swarms. A digital twin can act as the central hive mind, aggregating data from all robots and then sending back optimised instructions to each. For instance, in building a lunar structure, a swarm of 50 small rovers could continuously feed their position and status into the twin; the twin runs an algorithm to decide which rover should dig where or carry which block, and updates the plan in real-time as conditions change (such as one rover goes offline). This dynamic reconfiguration through the twin increases resilience, and even if several robots fail, the twin adapts the swarm's tasks so that construction continues. We already see early implementations: NASA has concepts for swarms assembling structures in orbit (e.g., swarms of free-flying bots building a large antenna), and Europe has done lab tests with micro-robots assembling blocks, guided by a central computer vision system (a form of twin). As swarm tech grows, standards might emerge for 'swarm-twin interfaces'. Additionally, robotics advancements such as soft robots or shape-shifting robots will provide new data to twins (imagine a flexible robot whose shape can change; its twin must incorporate a physics model for that flexibility). The twin will thus evolve to handle more complex robot kinematics and collaborative behaviours [47, 90].

## 6.3 | Extended Reality (XR) and Human-Twin Interaction

The way humans interact with digital twins will become more immersive and intuitive. We expect much greater use of Virtual

Reality (VR), Augmented Reality (AR), and Mixed Reality to visualise and control twins. In a VR environment, an engineer could virtually 'walk through' a lunar habitat's twin, inspecting systems as if present on the Moon. Already, companies such as Unity and NVIDIA's Omniverse are developing 3D platforms for digital twins that support VR viewing of factory twins or city twins. In construction, this means project stakeholders (even policymakers) could put on VR goggles and experience a remote project's status via its twin, improving communication and understanding. AR, on the other hand, will assist on-site crews. For example, a maintenance technician at a remote site could wear AR glasses that overlay the digital twin's info onto what they see, showing which buried cable segment exactly needs repair or highlighting an anomaly in a structure that is invisible to the naked eye but detected by the twin. In space, astronauts might rely on AR fed by the twin to perform repairs (there is talk of using Microsoft's HoloLens on the ISS for this type of guidance). XR also opens up multi-user collaboration in the twin—teams across the globe can meet 'inside' the digital twin model to discuss changes or issues, almost similar to a virtual conference at the construction site. This will be incredibly useful for international projects (e.g., a multinational Moon base programme) to make collective decisions efficiently [92, 93].

## 6.4 | Smart Materials and IoT Sensor Proliferation

The fidelity and utility of digital twins depend on data, which relies on sensors. We anticipate both an increase in the *quantity* of sensors (cheaper, smaller IoT devices scattered through structures) and the *quality* of sensors (new types that can measure things we could not before). Smart materials are materials that have built-in sensing or actuation. For example, concrete can sense its own strain or 'self-heal' cracks with embedded microcapsules. When such materials are used, their behaviour can be directly integrated into the twin. If a habitat is built from smart composite panels that change colour under stress (as a crude sensor), cameras can feed that info to the twin's structural model to indicate stress distribution. More advanced: self-healing materials might tell the twin how much of their healing capacity is left, so the twin knows when the material might need external intervention. By incorporating these, the twin becomes more accurate and can even simulate the *active* response of materials. A digital twin with smart material feedback could potentially initiate repairs, for example, send a current to a smart rebar grid to heat and heal cracks as soon as the twin detects micro-cracking. Thus, the line between the twin and the physical asset blurs as materials become part of the cyber-physical loop.

Furthermore, tiny energy-harvesting sensors could be sprinkled in places previously unmonitored (such as deep in a glacier supporting a structure or throughout a spacecraft's skin). This proliferation of IoT will feed the twin big data, which loops back to the need for AI to make sense of it. An exciting development is satellite IoT constellations, low earth orbit satellites that collect data from remote IoT sensors anywhere on Earth. This means even the most isolated construction site can pipe data to its twin in the cloud in near real-time without having to build local comms infrastructure. It is a synergy of space tech aiding terrestrial twins [65, 94].



## 6.5 | Federated and Scaled Digital Twin Ecosystems

We touched on connected twins, and the future likely holds not just individual twins, but large ecosystems or federations of twins. For example, a 'Mars colony twin' might consist of many sub-twins (life support, habitat structures, power grid, rover fleet) that all interface. Managing such complexity will require hierarchical or distributed twin architectures. Techniques such as agent-based modelling could allow each sub-twin to be an agent that communicates with others. The Gemini Principles in the UK foresee an 'ecosystem of connected digital twins' where data flows securely between domains. Imagine a scenario in 2040: multiple lunar bases by different countries exist; each has its digital twin. For global safety, these twins might share certain data (such as the location of all active rovers to avoid interference or environmental readings). Thus, an Internet of Digital Twins might emerge, analogous to today's Internet of computers. Policies and technology will need to allow selective, on-demand connectivity. For example, a twin can request info from another twin about a shared resource (such as water usage in a shared crater). Cloud platforms are already gearing up for this kind of thing, for example, Microsoft Azure and AWS have 'digital twin' services aiming to host multiple twin instances that can connect via APIs. On Earth, smart cities might link their twins for mutual benefit (a twin of a power plant linking with the city twin to manage distribution, etc.). All of this points to scale: from single assets to system-of-systems twins, covering entire operations or even regions [72, 96].

## 6.6 | Increased Autonomy and Closed-Loop Construction

Ultimately, the goal in very remote or dangerous environments is fully autonomous construction, machines building with minimal human intervention. Digital twins will serve as the central control system for such autonomy. A futuristic example: a crewless mission lands on Mars with robotic builders and supplies. A digital twin of the entire mission (hosted maybe on an orbiting satellite for computational power) orchestrates the construction of a habitat before humans arrive. It processes data, commands robots, and deals with surprises (similar to a dust storm) by re-planning, essentially acting as an AI site foreman. Although this level of autonomy might be a couple of decades out, the components are being put in place: AI planning, robust hardware, and high-fidelity simulations to validate actions. The twins would constantly simulate and verify the robots' work as they progressed, making corrections on the fly. Success in such an endeavour would rewrite construction manuals and would likely trickle back to Earth for automated construction in hazardous sites (imagine the cleanup of a radioactive site entirely by autonomous robots guided by a twin) [15, 97].

## 6.7 | Quantum Computing and High-Fidelity Physics

Looking even further, as quantum computing matures, it could vastly improve the simulation aspect of digital twins. Complex

physics (fluid dynamics, material quantum effects, etc.) that are too slow to simulate in real-time today might become tractable. This means twins could incorporate much more detailed physics, for example, precisely simulating how lunar dust will levitate and coat surfaces during rocket landings (a very complex plasma physics problem) to design better dust mitigation measures for habitats. High-fidelity twins may also allow real-time optimisation (such as shape changes to structures in response to loads). If smart materials allow structures to adapt, a quantum-enhanced twin could compute the optimal structural reconfiguration in real-time during, say, an earthquake, and instruct the structure to adjust accordingly. Although speculative, this hints at a future where the line between simulation and reality is extremely thin. The twin is so accurate and so tightly coupled that it essentially becomes the control system for the structure continuously [98, 99].

In summary, the future of digital twins in space and remote construction is deeply intertwined with other cutting-edge tech. AI will make twins smarter; robotics will make them more action-oriented; XR will make them more user-friendly; smart materials and IoT will make them richer in data; and connectivity will make them more collaborative. All these trends reinforce each other. Importantly, this future is not just about technology for its own sake. It serves a vision of creating infrastructure that is self-aware, self-optimising, and self-healing to the greatest extent possible. That vision is particularly pertinent in environments where human intervention is costly or impossible.

## 7 | Conclusion

Digital twin technology is transitioning from a novel concept to a practical necessity for tackling some of the most challenging construction and infrastructure projects faced by humanity. In extreme environments, whether the airless plains of the Moon, the dusty Martian horizon, the crushing depths of the ocean, or the frigid expanse of polar tundra, digital twins provide a critical bridge between the physical and the digital, between presence and remoteness, between the known and the unknown. This article has examined how digital twins function as high-fidelity virtual counterparts to physical systems, and how they are being applied to re-imagine construction and maintenance in environments where traditional approaches fall short. By enabling continuous monitoring, simulation-driven planning, and even remote control of assets, digital twins amplify our ability to build and sustain resilient infrastructure in places we cannot always be.

For multidisciplinary engineers and policymakers alike, the message is clear: embracing digital twins is a strategic move towards smarter, safer, and more sustainable infrastructure. The case studies from NASA, ESA, JAXA, SpaceX, and global industry leaders demonstrate that digital twins are already preventing failures, improving efficiencies, and saving costs in pilot projects around the world. The next step is scaling these successes through supportive policies, standardisation of best practices, and continued research and development. Policymakers have a role in incentivising the integration of digital twins into critical infrastructure and ensuring the frameworks

(legal, educational, technical) are in place to support them. When planning a lunar base or a new Arctic highway today, considering the digital twin is as important as considering the physical design; it should be baked into the project from inception as an essential component, not an afterthought.

Looking forward, as technology continues to advance, the synergy of digital twins with AI, robotics, XR, and novel materials will unlock capabilities that today might seem similar to science fiction, fully autonomous construction sites, ecosystems of interconnected twins optimising entire cities or planets, and perhaps even digital twins of humans (astronaut avatars) working side by side with physical robots. Each of these developments will further reduce the gap between what is imaginable and what is achievable in extreme environment construction. The transformative potential is immense: we will be able to 'live' in these distant or inhospitable environments virtually, solving problems and refining designs in real-time, and then confidently execute those solutions in the physical world.

In conclusion, digital twins are not just a tool but a paradigm shift in engineering and infrastructure management. They embody the convergence of the digital revolution with the age-old practice of building in the physical world. For extreme environment construction, this convergence offers a lifeline, a way to proceed with ambition but also with caution, informed by data and foresight. As we embark on building out the final frontier of space and securing the most remote corners of Earth, digital twins will be our reliable partners, illuminating the path, predicting the pitfalls, and helping us create structures that stand the test of time and nature. The journey is just beginning, but the trajectory is set: the future of construction, especially where it is hardest, will be co-created by engineers and their digital twins. By harnessing this technology wisely, we can turn bold visions, lunar habitats, underwater colonies, and resilient polar networks into reality, sustainably, and resiliently, for the benefit of all.

## Author Contributions

**Ehsan Noroozinejad Farsangi:** conceptualization, investigation, writing – original draft, methodology, visualization, writing – review and editing, formal analysis, project administration, data curation, supervision, resources.

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The data that support the findings of this study are available from the corresponding author upon reasonable request.

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